Final Paper Stage 2

Core paper-based forecast comparison

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4.1 Purpose

Stage 2 compares inflation forecasts using the corrected Stage 1 data, with every series taken as the end-of-quarter value of the monthly FRED series and inflation computed as $\pi_t = 400 \log(P_t/P_{t-1})$ from CPIAUCSL. We check stationarity over 1960Q1–1999Q4 (4.2), set up an expanding-window experiment (4.3), build the seven core models from Faust and Wright (2013) (4.4), and compare their forecasts through 2019Q4 (4.5). Lastly, we diagnose the best one-step model (4.6) and summarize the results (4.7).

4.2 Assess stationarity and structural change

We use the initial period, 1960Q1–1999Q4 (160 quarters), for every test in this section. Figure 1 shows the three series, with the segment means from a change-point analysis of inflation described below, and Figure 2 shows their autocorrelation functions.

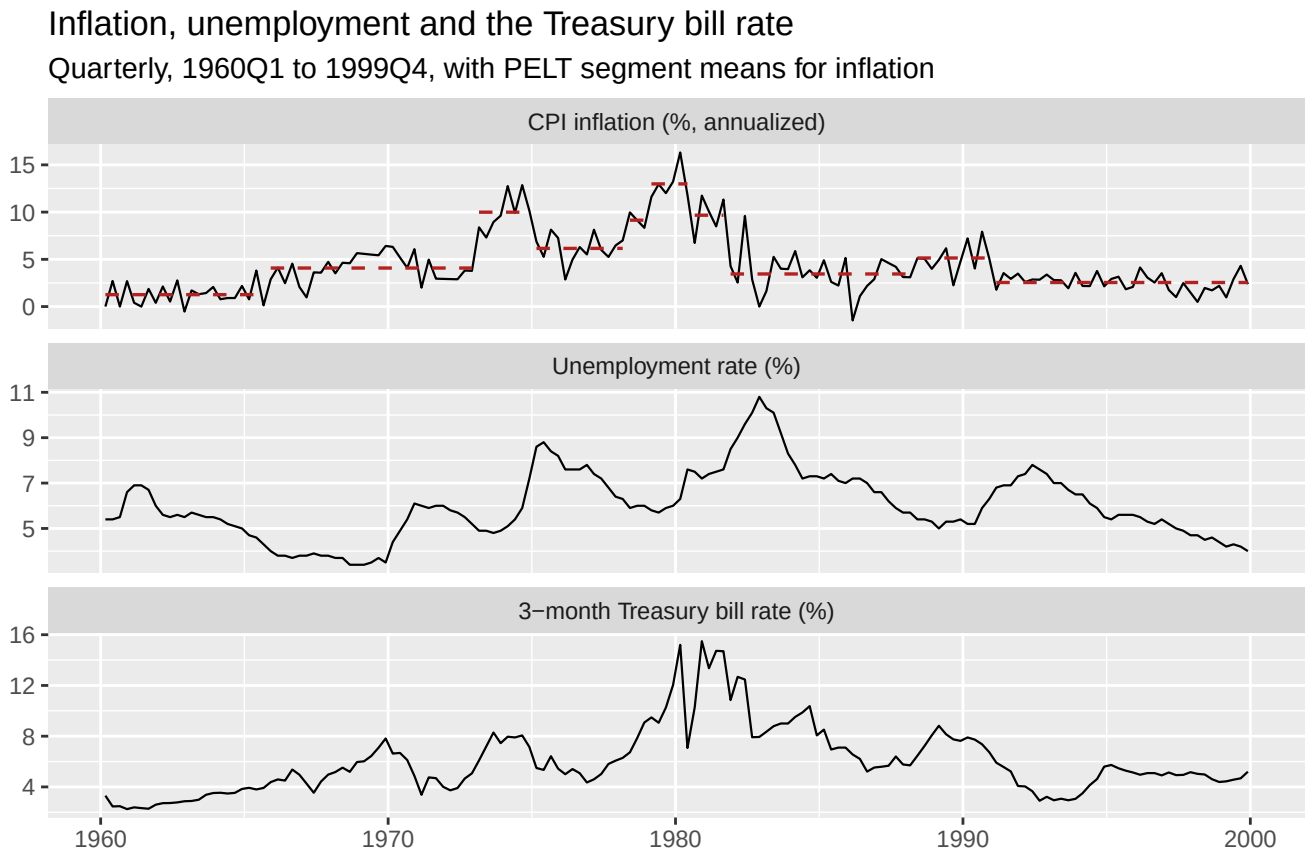


Figure 1: The three series, 1960Q1–1999Q4. Dashed red lines in the inflation panel are PELT segment means after rescaling inflation by its noise scale.

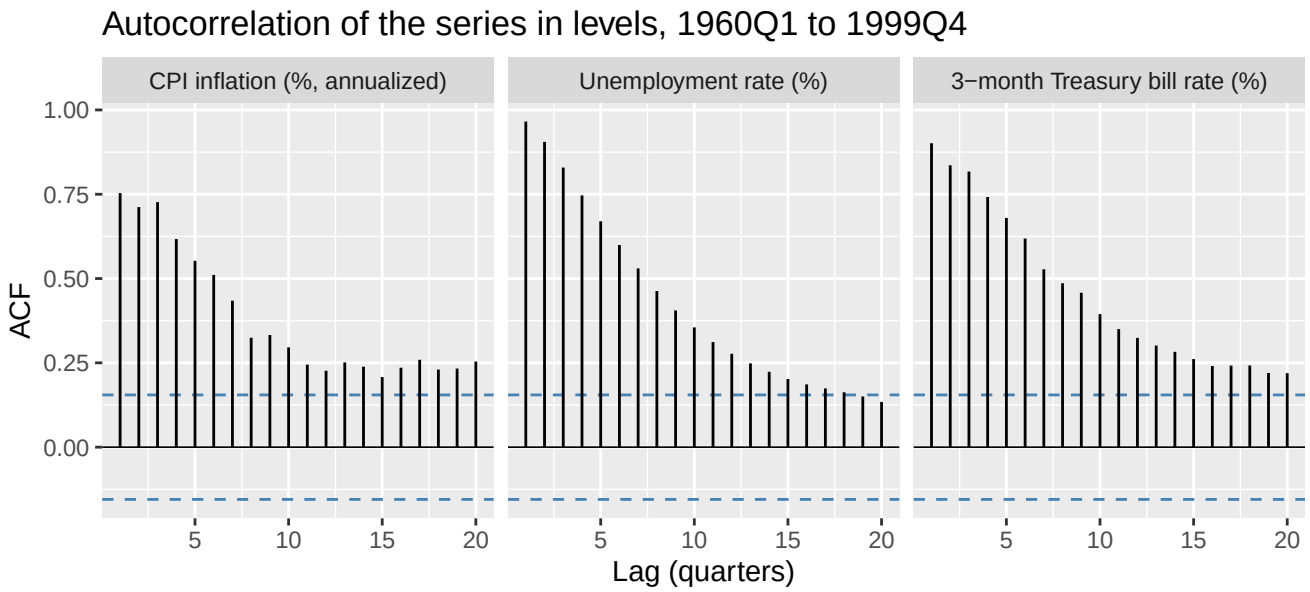


Figure 2: Sample ACFs of the three series in levels, lags 1–20. Dashed lines are approximate 95% bounds.

What the plots show. Inflation rises from about 1% in the early 1960s to double digits in 1974 and 1979–80, then falls back after 1981. Its ACF starts at 0.75 and is still 0.32 at lag 8, so dependence is persistent, but it decays steadily rather than staying near one. Unemployment and the Treasury bill rate are more persistent still, with lag-1 autocorrelations of 0.97 and 0.90. Unemployment moves in long cyclical swings, and the bill rate tracks inflation upward before spiking during the 1979–82 Volcker tightening. None of the three series returns quickly to a single fixed level.

Unit-root tests. For each series we run the augmented Dickey–Fuller (ADF) test with a constant and no trend, since none of these rates has an economic reason to trend deterministically. We choose the augmentation lag from 0–4 by BIC, comparing all five regressions on the same estimation sample so the BIC values are comparable, and then run `urca::ur.df()` at the selected lag. We don't use the `selectlags = "BIC"` option in `ur.df()` here, since it only searches lags 1 through the maximum and can't select 0. The KPSS test uses a constant (`type = "mu"`) and the short Newey–West bandwidth of 4 lags, and we repeat it with the long bandwidth of 13 lags as a robustness check. The ADF null hypothesis is a unit root, and the KPSS null hypothesis is stationarity. When the level results support a unit root, we repeat both tests on the first difference. Table 1 collects the results.

In levels, no ADF statistic reaches the 5% critical value of -2.88, and every short-bandwidth KPSS statistic exceeds its 5% critical value of 0.463, though none reaches the 1% value of 0.739. In first differences, both tests indicate stationarity for all three series. The KPSS rejections are also sensitive to the bandwidth. With the long bandwidth, the level statistics fall to 0.235, 0.250 and 0.289, and none reject. The long bandwidth trades power for size, so this doesn't show the series are stationary, but it does mean the KPSS evidence against stationarity is weak for every series.

Structural change. Applied to raw inflation, PELT finds the 17 breaks reported in Stage 1, about one every 9 quarters, which looks more like PELT tracking a wandering series than like a few regimes. The reason is that `cpt.mean()` assumes unit noise variance, while inflation's quarter-to-quarter noise has a standard deviation of about 1.43 points, so the BIC penalty is too small for the scale of the data. We therefore divide inflation by a robust estimate of its noise scale, $\mathrm{MAD}(\Delta\pi_t)/\sqrt{2}$, before running PELT, which changes the Stage 1 settings. The rescaled analysis finds 9 breaks (Figure 1). They describe inflation near 1.3% through 1965, about 4.1% through 1972, a high and volatile 1973–81 with segment means between 6.2% and 13.0%, and a lower level after 1981, with segment means of 3.4%, 5.1% and 2.5%. These shifts matter for the tests, since a series that is stationary around a mean that shifts a few times can make the ADF test fail to reject and the KPSS test reject (Perron 1989), which is exactly the pattern in levels above. Ignoring the shifts could lead us to treat inflation as having a unit root when it may instead be reverting to a slowly changing level. We don't use PELT for unemployment or the bill rate. Their quarter-to-quarter changes are small relative to their slow swings, so even after the same rescaling PELT splits them into 28 to 31 segments, which describes persistence rather than separate regimes.

Table 1: Stationarity summary, 1960Q1–1999Q4. ADF entries are τ statistics with the BIC-selected lag, and KPSS entries are test statistics with the short bandwidth unless marked long. 5% critical values are -2.88 (ADF) and 0.463 (KPSS).

Series	ADF evidence	KPSS evidence	Structural-change evidence	Classification
CPI inflation	Level -2.27, lag 2 (do not reject). Difference -15.24, lag 1 (reject).	Level 0.500 (reject), long 0.235 (do not reject). Difference 0.075 (do not reject).	9 mean breaks after rescaling. Low early 1960s, high 1973–81, lower after 1981.	Unresolved
Unemployment	Level -2.38, lag 1 (do not reject). Difference -7.41, lag 0 (reject).	Level 0.541 (reject), long 0.250 (do not reject). Difference 0.098 (do not reject).	No discrete regimes. Long cyclical swings.	$I(1)$
Treasury bill rate	Level -2.44, lag 3 (do not reject). Difference -7.23, lag 2 (reject).	Level 0.646 (reject), long 0.289 (do not reject). Difference 0.075 (do not reject).	No discrete regimes. Rises with inflation, 1979–82 spike.	$I(1)$

Classification. We classify unemployment and the Treasury bill rate as $I(1)$. The ADF test never rejects, differencing removes the persistence, and their lag-1 autocorrelations of 0.97 and 0.90 decay only slowly, so a unit root is the simpler description even though the KPSS evidence is weak. Both rates are bounded in the long run, so we read $I(1)$ as a description of their behavior over this sample rather than a claim about their true process. We classify inflation as unresolved. Its unit-root tests look like those of the two rates, but its persistence is much lower, with a lag-1 autocorrelation of 0.75 that falls to 0.32 by lag 8, and its mean shifts at identifiable dates that group into three broad regimes, a low early 1960s, a high 1973–81 and a lower level after 1981. A series that is stationary around a shifting mean produces the same test results, while the two rates move too persistently for a few mean shifts to explain them. That is why the AR models in 4.4.3 and 4.4.4 are worth comparing against the gap models in 4.4.5, since the AR models impose one fixed long-run mean and the gap models track a local one. The forecasts will show which description of inflation works better out of sample.

4.3 Set up the expanding-window forecast experiment

A forecast origin T is the quarter a forecast is made in and the last quarter of its estimation sample. The first origin is 1999Q4, so every model is first fit on 1960Q1–1999Q4 and forecasts $h = 1$, $h = 4$, and $h = 8$ quarters ahead. We then add one quarter, refit, and forecast again until the targets reach 2019Q4. Lag selection and the local mean are redone inside each origin's sample too, so nothing after T is ever used.

Forecast origin	Data used at that origin	$h = 1$ target	$h = 4$ target	$h = 8$ target
1999Q4	1960Q1–1999Q4	2000Q1	2000Q4	2001Q4
2000Q1	1960Q1–2000Q1	2000Q2	2001Q1	2002Q1

Table 2: Forecast origins, targets, and number of errors by horizon.

Horizon	First origin	Last origin	Target period	Number of errors
$h = 1$	1999Q4	2019Q3	2000Q1–2019Q4	80
$h = 4$	1999Q4	2018Q4	2000Q4–2019Q4	77
$h = 8$	1999Q4	2017Q4	2001Q4–2019Q4	73

The forecast error is

$$e_{j,t}^{(h)} = \pi_t - \hat{\pi}_{j,t|t-h}^{(h)},$$

so a positive error means the model underpredicted. This gives 80, 77, and 73 errors per model at $h = 1$, $h = 4$, and $h = 8$ (Table 2), since longer horizons run out of targets sooner.

Sources: [W4S2P15, W4S2P16, W4S2P18]

4.4 Implement the core forecasting models

We compare seven models. The random walk and four-quarter average are benchmarks with nothing to estimate, and the other five forecast from estimated or constructed relationships. Each one is written as `fc_<model>(data, H)`, so the framework in 4.3 runs all seven the same way.

4.4.1 Random-walk model

The random walk without drift treats each quarter's inflation as last quarter's inflation plus an unpredictable shock:

$$\pi_t = \pi_{t-1} + \varepsilon_t, \qquad E(\varepsilon_t) = 0, \qquad Var(\varepsilon_t) = \sigma_\varepsilon^2.$$

Since future shocks have mean zero, the best forecast at origin T is the most recent observation at every horizon:

$$\hat{\pi}_{T+h|T}^{RW} = \pi_T, \qquad h = 1, \ldots, 8.$$

In code, `fc_rw(data, H)` returns `data$Inflation[nrow(data)]` for every horizon H, using only the sample through T. The model has no parameters to estimate, so "re-estimating" it at each origin just means reading off the latest inflation rate, which makes it the simplest reference in the comparison. It never reverts toward any mean, which is where it parts company with a stationary AR, and it reacts fully to whatever happened in the last quarter, including noise.

4.4.2 Four-quarter-average benchmark

The four-quarter-average benchmark is used as a benchmark to be beaten by more complex models. In this naive method, the four most recently observed quarters are averaged to smooth out the noise in any one quarter, and this average is used as the forecast at every horizon. This estimate can be mathematically represented as

$$\hat{\pi}_{T+h|T}^{4Q} = \frac{1}{4} \sum_{j=0}^{3} \pi_{T-j},$$

where T is the most recently observed quarter and h is the forecast horizon. In code, `fc_4q(data, H)` returns the mean of the last four inflation values in the origin's sample at every horizon. Its RMSE is the denominator for every relative RMSE in 4.5.

4.4.3 AR(p) model

The AR(p) model forecasts inflation using its own past values. Unlike the four-quarter average, which gives the same forecast at every horizon, the AR(p) model uses current and lagged inflation to make forecasts that change with the horizon. At each forecast origin, we estimate autoregressive models with lag orders $p \in \{1, 2, 3, 4\}$ by maximum likelihood and select the model with the lowest Bayesian Information Criterion (BIC). The model is specified as

$$\pi_t = c + \sum_{j=1}^{p} \phi_j \pi_{t-j} + \varepsilon_t,$$

where c is the intercept, ϕ_j are the autoregressive coefficients, and ε_t is the error term. All four candidates are fit by exact maximum likelihood on the full sample through the origin, so their BIC values are comparable. Multistep forecasts are generated iteratively using the same estimated coefficients. If the model is stationary, forecasts converge toward its implied long-run mean. In code, `fc_arp(data, H)` fits the four candidates with `arima()`, keeps the one with the lowest BIC and returns its H-step forecast from `predict()`.

4.4.4 Direct AR model

The direct AR fits a separate regression for each horizon instead of iterating one equation forward.

$$\pi_{t+h} = c_h + \sum_{j=0}^{p_h-1} \phi_{h,j}\, \pi_{t-j} + \varepsilon_{t+h}.$$

At origin T, we only use rows with $t + h \leq T$, so the response is already observed. We pick p_h from 1 to 4 by BIC on a common set of rows, and the forecast plugs $\pi_T, \ldots, \pi_{T-p_h+1}$ into the chosen equation. BIC picks $p_h = 3$ at every origin for $h = 1$, between 1 and 3 for $h = 4$, and $p_h = 1$ at every origin for $h = 8$.

Sources: [W4S1P23, W4S1P24, W4S2P16, W4S2P18, W6S2P12, W7S2P33]

4.4.5 Local mean by exponential smoothing

Following Faust and Wright (2013), we estimate a slowly changing local mean τ_t by simple exponential smoothing with $\alpha = 0.05$, a weight the project fixes rather than estimates. We start the recursion at the first observation and update one quarter at a time:

$$\tau_{1960Q1} = \pi_{1960Q1}, \qquad \tau_t = 0.95\,\tau_{t-1} + 0.05\,\pi_t, \qquad g_t = \pi_t - \tau_t.$$

Each new quarter moves the local mean only 5% of the way toward current inflation, so τ_t follows sustained changes in the level but largely ignores one-quarter spikes. Unrolling the recursion shows that τ_t is a weighted average of current and past inflation with weights $0.05(0.95)^j$, which halve roughly every 13.5 quarters. The starting value $\tau_{1960Q1} = \pi_{1960Q1}$ is a single noisy observation, but its weight in τ_t is 0.95^{t-1}, which is 0.13 by 1970Q1 and 0.0003 by the first forecast origin, so the initialization has no practical effect on the local mean during the forecast period. The gap g_t measures how far inflation sits above or below that local level.

The recursion is one-sided because τ_t depends only on π_t and earlier values. At each forecast origin T, `add_local_mean()` runs the update from 1960Q1 through T on the origin's own sample, so τ_T and g_T use nothing after T. As a check, the local mean computed from data through 1999Q4 matches the 1960Q1–1999Q4 portion of the local mean computed through 2019Q4 exactly. The function also reproduces the guide's example, where a previous local mean of 3.0 and current inflation of 5.0 give $\tau_t = 3.1$ and $g_t = 1.9$.

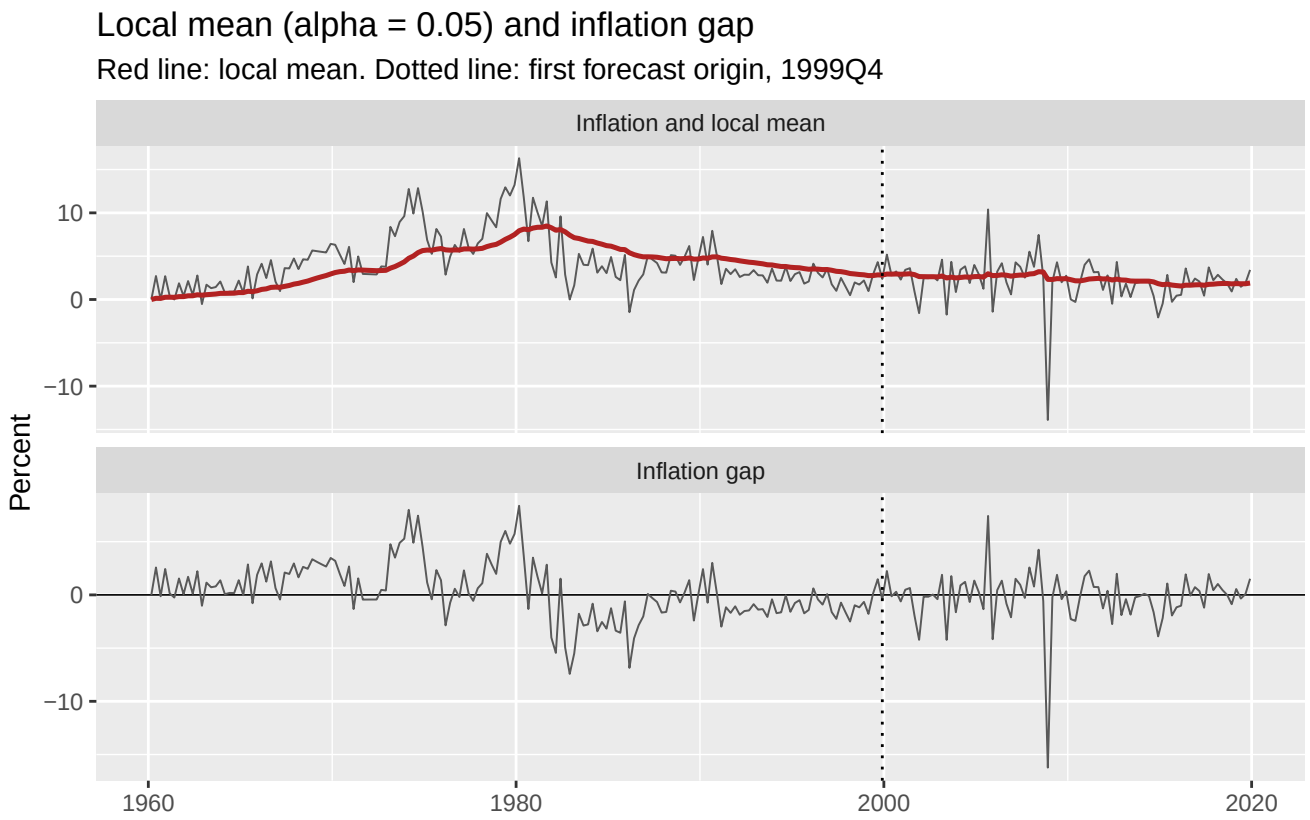


Figure 3: Inflation and its exponentially smoothed local mean (top), and the inflation gap (bottom), 1960Q1–2019Q4. Values through any origin T are the ones available at T.

Figure 3 shows how the local mean behaves. It climbs from 0.8% in 1965Q1 to a peak of 8.5% in 1981Q3, then declines slowly to 2.8% at the first forecast origin and 1.9% by 2019Q4. It lags turning points by several years, since the weight on new data is small. For example, the gap is 8.4 points in 1980Q1, when inflation surged well above the slowly rising local mean, and the local mean is still above inflation for much of the 1980s. Through the 1990s inflation is lower and steadier than the local mean, so the gap is mostly negative while the local mean drifts down to meet it. By the 2000s the gap is centered close to zero, with an average of -0.22 over 2000Q1–2019Q4. The gap is also less persistent than inflation itself. Its lag-1 autocorrelation over 1960Q1–1999Q4 is 0.66, compared with 0.75 for inflation.

The gap AR (4.4.6) and the fixed-persistence forecast (4.4.7) call `add_local_mean()` on each origin's sample and read τ_T and g_T from the last row. Both forecasts hold the local mean at τ_T, since the future inflation needed to update it isn't observed yet.

4.4.6 Gap AR model

The gap AR model forecasts inflation in two parts: the slowly changing local mean τ_t from 4.4.5, and the inflation gap $g_t = \pi_t - \tau_t$ around it. At each forecast origin, we compute τ_t and g_t from the origin's own sample and fit an autoregression with an intercept to the gap,

$$g_t = a + \sum_{j=1}^{p} \rho_j g_{t-j} + v_t,$$

by OLS for each $p \in \{1, 2, 3, 4\}$, keeping the lag order with the lowest BIC. All four regressions use the same estimation sample, starting in the fifth quarter, so their BIC values are comparable. We then forecast the gap iteratively, feeding each forecast back in as a lag,

$$\hat{g}_{T+k|T} = \hat{a} + \sum_{j=1}^{p} \hat{\rho}_j \, \hat{g}_{T+k-j|T}, \qquad k = 1, \ldots, h,$$

where $\hat{g}_{T+k-j|T} = g_{T+k-j}$ whenever $k - j \leq 0$, and add the current local mean back:

$$\hat{\pi}^{GAP}_{T+h|T} = \tau_T + \hat{g}_{T+h|T}.$$

The local mean is held at τ_T over the forecast period, since the future inflation needed to update it isn't observed yet. Comparing this model with the AR(p) model shows whether recentering inflation around a slowly changing level improves forecasts. In code, `fc_gap_ar(data, H)` runs `add_local_mean()` from 4.4.5 on the origin's sample, fits the four gap autoregressions, and iterates the chosen one forward H quarters.

4.4.7 Fixed-persistence gap forecast

The fixed-persistence gap model forecasts inflation using the slowly changing local mean τ_t from 4.4.5 and the inflation gap $g_t = \pi_t - \tau_t$. Following Faust and Wright (2013), we fix the persistence coefficient at 0.46 rather than estimating it at each forecast origin. The forecast is specified as

$$\hat{\pi}^{FIX}_{T+h|T} = \tau_T + 0.46^h g_T,$$

where T is the most recently observed quarter and h is the forecast horizon. At each expanding-window origin, we calculate the local mean and gap using only the available observations. The local mean is held at τ_T over the forecast period, while the inflation gap shrinks geometrically toward zero as the horizon increases. Comparing this model with the estimated gap AR model shows whether estimating additional persistence parameters improves forecast accuracy enough to offset their estimation uncertainty. In code, `fc_fixed_gap(data, H)` runs `add_local_mean()` on the origin's sample and returns $\tau_T + 0.46^H g_T$.

4.4.8 Direct Phillips-curve model

The direct Phillips curve is the direct AR with the unemployment rate at t added.

$$\pi_{t+h} = c_h + \sum_{j=0}^{p_h - 1} \phi_{h,j} \, \pi_{t-j} + \lambda_h u_t + \varepsilon_{t+h}.$$

We keep u_t in every equation, pick the inflation lags by BIC the same way, and forecast with u_T, which is already observed at the origin. The unemployment coefficient averages -0.178 at $h = 1$, -0.145 at $h = 4$, and 0.091 at $h = 8$. The negative sign at the shorter horizons fits the Phillips curve, but the sign flips at $h = 8$, so unemployment likely doesn't say much about inflation that far out.

Sources: [W7S2P6, W7S2P9, W7S2P33, W9S2P30]

4.5 Evaluate the forecasts

We run all seven models through the same expanding window and compute the following for each model j and horizon h.

$$RMSE_{j,h} = \sqrt{\frac{1}{n_h}\sum_t \left(e_{j,t}^{(h)}\right)^2}, \qquad rRMSE_{j,h} = \frac{RMSE_{j,h}}{RMSE_{4Q,h}},$$

$$MAE_{j,h} = \frac{1}{n_h}\sum_t \left|e_{j,t}^{(h)}\right|, \qquad ME_{j,h} = \frac{1}{n_h}\sum_t e_{j,t}^{(h)}.$$

RMSE is the main ranking measure, and a relative RMSE below 1 means the model beats the four-quarter average.

Sources: [W4S2P8, W4S2P9, W4S2P10, W4S2P18]

4.5.1 Random-walk forecasts

Table 3: Random-walk forecast accuracy, targets through 2019Q4.

Horizon	Target period	n_h	RMSE	Relative RMSE	MAE	Mean error
1	2000Q1–2019Q4	80	3.82	1.26	2.44	0.01
4	2000Q4–2019Q4	77	3.94	1.35	2.49	-0.06
8	2001Q4–2019Q4	73	3.83	1.25	2.55	-0.09

We generate the random-walk forecasts at every origin from 1999Q4 to 2019Q3, keep $h = 1, 4, 8$, and evaluate them against targets through 2019Q4, which gives 80, 77 and 73 forecast errors, matching the counts in the guide. The relative RMSE below uses the 4.4.2 benchmark forecasts from the same origins.

Relative RMSE against the four-quarter-average benchmark is 1.26 at $h = 1$, 1.35 at $h = 4$ and 1.25 at $h = 8$, so the random walk's RMSE runs 25 to 35 percent above the benchmark's, with the gap widest at $h = 4$. In levels, RMSE is 3.82 at $h = 1$, 3.94 at $h = 4$ and 3.83 at $h = 8$, so accuracy barely changes with the horizon. The random walk carries the full noise of π_T into every forecast, so even a one-quarter-ahead forecast inherits the volatility of the latest observation. The largest one-step error is -16.5 points, for 2008Q4, when the forecast carried forward 2008Q3 inflation of 2.6% and actual inflation was -13.9%. RMSE runs well above MAE at every horizon, 3.82 against 2.44 at $h = 1$, so a few very large errors drive much of the RMSE. Targets in 2008 and 2009 account for 50% of the squared one-step error, even though they're only 8 of the 80 quarters. Mean errors are close to zero (0.01, -0.06 and -0.09), but for a random walk this is mechanical rather than evidence of no bias. Each error is the change in inflation from the origin to the target, so the errors telescope. At $h = 1$ their mean is $(\pi_{2019Q4} - \pi_{1999Q4})/80 = (3.42 - 2.38)/80 \approx 0.01$, and at $h = 4$ and $h = 8$ the same cancellation leaves only the last h targets and the first h origins. A mean error near zero only says that inflation ended the evaluation sample close to where it started.

A random walk's h-step forecast error is the sum of h independent shocks, so its RMSE should grow roughly as $\sqrt{h}$. From 3.82 at $h = 1$, that implies about 7.64 at $h = 4$ and 10.81 at $h = 8$, far above the 3.94 and 3.83 we observe. The comparison isn't exact, since the targets overlap across origins and the 2008–09 errors reach every horizon, but dropping the 2008 and 2009 targets still leaves RMSE at 2.86, 2.73 and 3.58, nowhere near $\sqrt{h}$ growth. The variance of the h-quarter change in inflation levels off rather than accumulating, so most quarter-to-quarter movement is transitory, which rules out a pure random walk. It doesn't settle the $I(0)$ versus $I(1)$ question from 4.2, though, since a slowly drifting $I(1)$ level plus large temporary noise produces the same flat profile. The local mean and gap in 4.4.5 separate exactly those two components, so the test of this reading is how the gap AR (4.4.6) and fixed-persistence forecasts (4.4.7) compare with the random walk and the AR models.

4.5.2 Direct AR and Phillips-curve forecasts

Table 4: Direct AR and direct Phillips-curve forecast accuracy, targets through 2019Q4.

Model	Horizon	Target period	n_h	RMSE	Relative RMSE	MAE	Mean error
Direct AR	1	2000Q1–2019Q4	80	3.09	1.02	1.92	-0.32
Direct AR	4	2000Q4–2019Q4	77	3.10	1.06	2.00	-0.78
Direct AR	8	2001Q4–2019Q4	73	3.24	1.06	2.18	-1.47
Direct Phillips curve	1	2000Q1–2019Q4	80	3.15	1.04	1.99	-0.34
Direct Phillips curve	4	2000Q4–2019Q4	77	3.11	1.06	2.01	-0.78
Direct Phillips curve	8	2001Q4–2019Q4	73	3.29	1.08	2.26	-1.58

Neither direct model beats the four-quarter average at any horizon. The direct AR's relative RMSE is 1.02, 1.06, and 1.06 at $h = 1$, $h = 4$, and $h = 8$, and the Phillips curve's is 1.04, 1.06, and 1.08, so adding unemployment doesn't help. Both models also overpredict, and the direct AR's mean error grows from -0.32 at $h = 1$ to -1.47 at $h = 8$. This likely comes from the long-run level each equation implies, which is estimated mostly from the higher inflation before 2000.

Sources: [W4S1P23, W4S2P8, W4S2P9, W4S2P10, W4S2P15, W4S2P18]

4.5.3 All seven models

Table 5: Expanding-window forecast accuracy for all seven models, targets through 2019Q4. Relative RMSE is against the four-quarter average. Models are sorted by RMSE within each horizon.

Model	Horizon	n_h	RMSE	Relative RMSE	MAE	Mean error
Fixed-persistence gap	1	80	3.00	0.99	1.87	-0.12
Four-quarter average	1	80	3.04	1.00	1.87	-0.01
Gap AR	1	80	3.06	1.01	1.88	-0.09
AR(p)	1	80	3.08	1.01	1.91	-0.27
Direct AR	1	80	3.09	1.02	1.92	-0.32
Direct Phillips curve	1	80	3.15	1.04	1.99	-0.34
Random walk	1	80	3.82	1.26	2.44	0.01
Fixed-persistence gap	4	77	2.73	0.93	1.62	-0.30
Gap AR	4	77	2.86	0.98	1.72	-0.26
Four-quarter average	4	77	2.92	1.00	1.75	-0.09
AR(p)	4	77	2.94	1.01	1.81	-0.62
Direct AR	4	77	3.10	1.06	2.00	-0.78
Direct Phillips curve	4	77	3.11	1.06	2.01	-0.78
Random walk	4	77	3.94	1.35	2.49	-0.06
Fixed-persistence gap	8	73	2.78	0.91	1.69	-0.37
Gap AR	8	73	2.89	0.95	1.86	-0.40
Four-quarter average	8	73	3.05	1.00	2.04	-0.11
AR(p)	8	73	3.06	1.00	2.05	-0.95
Direct AR	8	73	3.24	1.06	2.18	-1.47
Direct Phillips curve	8	73	3.29	1.08	2.26	-1.58
Random walk	8	73	3.83	1.25	2.55	-0.09

The fixed-persistence gap has the lowest RMSE at all three horizons (3.00, 2.73, and 2.78), and its edge over the four-quarter average grows with the horizon (1.2%, 6.7%, and 8.8%). The random walk is the worst at every horizon, with relative RMSE from 1.25 to 1.35. Since fixing the gap's persistence at 0.46 beats estimating it in the gap AR, the gap AR's extra parameters likely add more noise than they gain in fit.

4.6 Diagnose the leading one-step model

Among the estimated models, the fixed-persistence gap has the lowest $h = 1$ RMSE (3.00), so it's the one we diagnose. Refitting it through 2019Q4 gives 239 one-step residuals, $\pi_t - (\tau_{t-1} + 0.46\, g_{t-1})$, from 1960Q2 to 2019Q4.

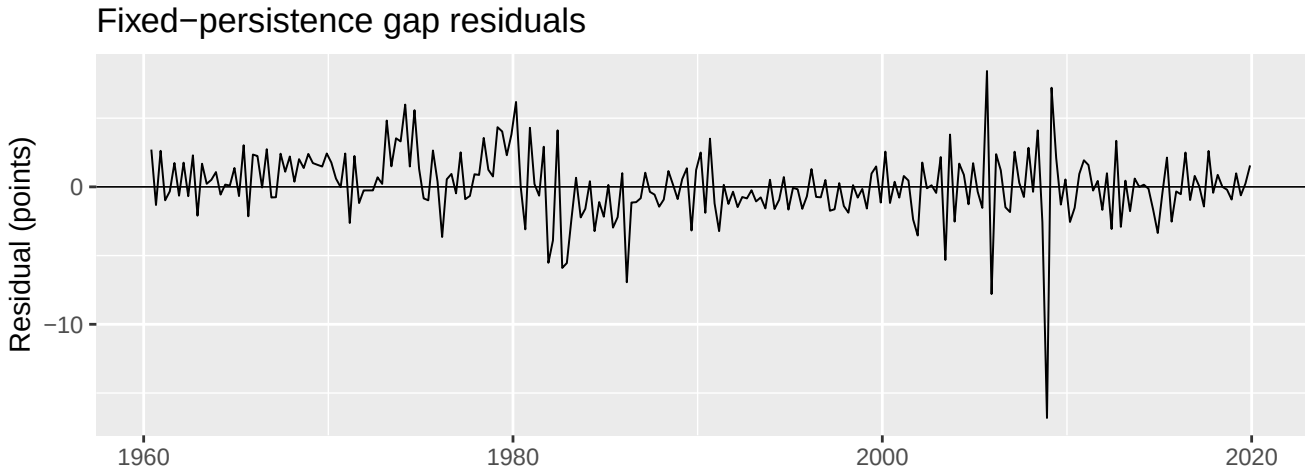


Figure 4: Fixed-persistence gap one-step residuals, 1960Q2–2019Q4.

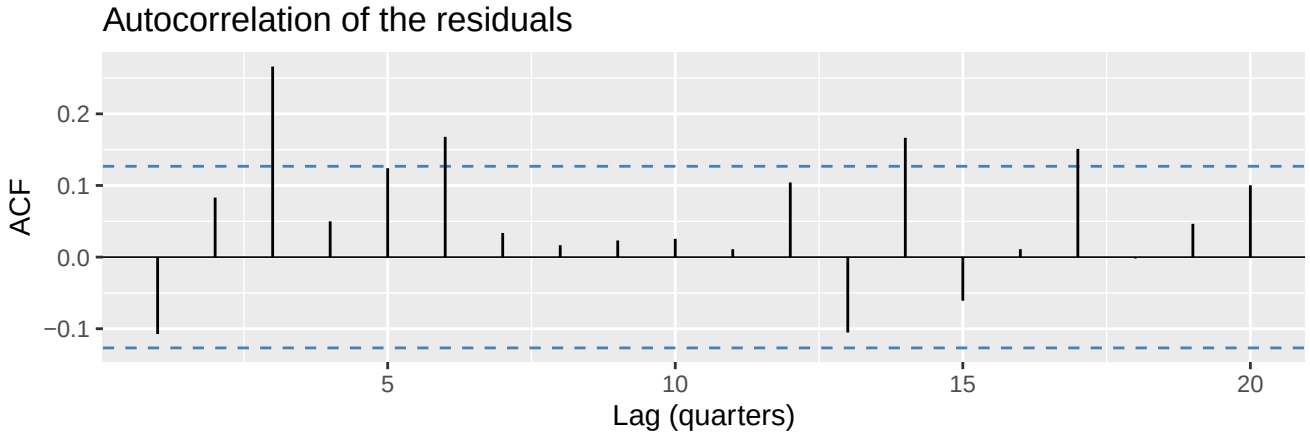


Figure 5: Sample ACF of the residuals, lags 1–20. Dashed lines are approximate 95% bounds.

The residuals are centered near zero, but their spread clearly isn't constant, with the biggest swings in the 1970s, early 1980s, and 2000s and the largest residual in 2008Q4. The ACF falls outside the 95% bounds at lags 3, 6, 14, and 17. Following the course rule $\ell = \min(10, T/5) = 10$, the Ljung–Box statistic is 33.76 with a p-value below 0.001, so we reject the null of no autocorrelation. This means the fixed persistence of 0.46 doesn't capture all of the dependence in inflation.

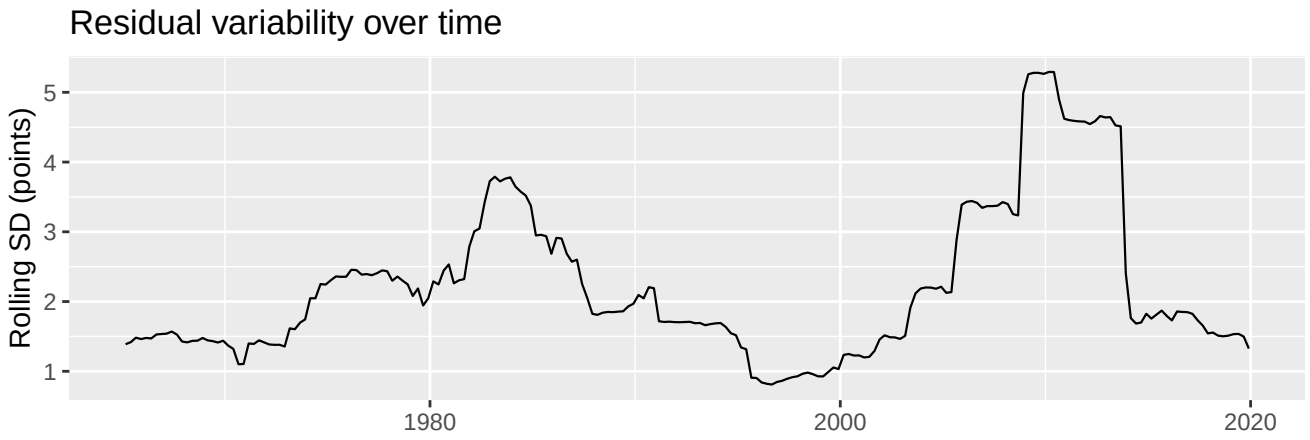


Figure 6: 20-quarter rolling standard deviation of the residuals. Each point covers the 20 quarters ending in that quarter.

Table 6: The five largest one-step residuals, 1960Q2–2019Q4.

Quarter	Inflation	Fitted	Residual	Squared residual
2008Q4	-13.91	2.91	-16.82	282.8
2005Q3	10.40	1.97	8.43	71.1
2005Q4	-1.41	6.39	-7.80	60.8
2009Q1	2.07	-5.14	7.21	52.0
1986Q1	-1.46	5.48	-6.94	48.2

The 20-quarter rolling standard deviation ranges from 0.81 to 5.29, so the error variance changes over time and intervals built from a single variance wouldn't be reliable. The five largest residuals (Table 6) also make up 35.0% of the sum of squared residuals, so a few quarters like 2008Q4 could decide the DM–HLN comparisons in the final project.

Sources: [W6S2P21, W6S2P23]

4.7 Summarize the Stage 2 results

The fixed-persistence gap has the lowest RMSE at every horizon and beats the four-quarter average at all three, while the gap AR also beats it at $h = 4$ and $h = 8$. The random walk is the worst at every horizon. MAE is only 59 to 69% of RMSE, so a few big misses like 2008Q4 push RMSE up for every model. Lastly, 20 of the 21 model and horizon combinations have negative mean error, so overprediction is the norm from 2000 to 2019. The DM–HLN tests come in the final project.

Sources: [W4S1P23, W4S1P24, W4S2P8, W4S2P9, W4S2P10, W4S2P18, W4S2P19]

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